Available online at www.sciencedirect.com
ScienceDirect
Energy Reports 8 (2022) 748–755
www.elsevier.com/locate/egyr

2022 International Conference on Energy Storage Technology and Power Systems (ESPS 2022), February 25–27, 2022, Guilin, China

Data-driven alternating current optimal power flow: A Lagrange multiplier based approach

Xingyu Lei[a,*], Juan Yu[a], Habaer Aini[b], Wencui Wu[c]

[a] State Key Laboratory of Power Transmission Equipment & System Security and New Technology, Chongqing University, Chongqing 400044, China
[b] Urumqi Power Supply Company, Xinjiang 830000, China
[c] Bortala Power Supply Company, Xinjiang 834000, China

Received 1 October 2022; accepted 3 October 2022
Available online 18 October 2022

Abstract

This paper proposes a data-driven Alternating Current Optimal Power Flow (AC-OPF) method assisted by Lagrange multipliers. Stacked Extreme Learning Machine (SELM) is introduced for AC-OPF learning to avoid the time-consuming training and hyperparameter adjustment process of deep neural networks. Instead of incorporating the prior physical information into the neural networks algorithm, we developed a new neural network structure for the SELM learning based on the Lagrange multipliers of the AC-OPF problem. Case studies of several IEEE benchmark systems demonstrate that the AC-OPF learning performance is improved by introducing additional Lagrange multipliers information, while the proposed method outperforms other alternatives with almost 99% learning accuracy and fast computation.

Peer-review under responsibility of the scientific committee of the International Conference on Energy Storage Technology and Power Systems, ESPS, 2022.

Keywords: Alternating current optimal power flow; Data-driven; Lagrange multipliers; Stacked extreme learning machine

1. Introduction

As a fundamental tool of power system operation, the alternating current optimal power flow (AC-OPF) is generally challenging to solve in real-time. Traditionally, the AC-OPF problem is solved based on the physical model, which needs numerous time-consuming iterations, leading to a high computational cost for calculation [1,2]. In a high renewable energy-connected power system, the AC-OPF problem has to be solved more frequently to maintain the optimal system operation in real-time. This need has paved the way for the new AC-OPF methods with a low computational cost for the high renewable energy-connected power system.

In recent years, regarding the excellent learning performance of deep neural networks (DNNs), several data-driven approaches based on DNNs have been applied in power system operation. It is a promising tool to solve

* Corresponding author.
E-mail address: lxylxy7@163.com (X. Lei).

https://doi.org/10.1016/j.egyr.2022.10.024

Peer-review under responsibility of the scientific committee of the International Conference on Energy Storage Technology and Power Systems, ESPS, 2022.

the AC-OPF problem with fast computational speed and high precision. In [3–5], DNNs are trained for the OPF calculation are developed based on offline simulations and historical data. The key idea of these approaches is to construct a neural network that learns the AC-OPF mapping relationship between the power injection and optimal solutions, thus significantly improving the AC-OPF computational efficiency. To enhance the learning efficiency of DNNs, several studies suggest incorporating the prior physical information into the DNNs algorithm. For example, Refs. [6,7] propose to train a DNN to learn the AC-OPF mapping relationship with the minimization of training loss with augmented Lagrangian relaxation. Ref. [8] combines DNNs and Lagrangian duality to learn the AC-OPF solutions, which reported training process speedup. However, a mass of hyperparameters adjustment is needed for these DNN-based methods. The learning ability of a DNN-based method depends mainly on its hyperparameter adjustment, but no efficient algorithm is developed to guide this process.

This paper develops a data-driven AC-OPF method. Stacked Extreme Learning Machine (SELM) in introduced to overcome the hyperparameter adjustment issue of DNNs. Compared with DNNs, SELM does not require the time-consuming hyperparameter adjustment process and has a fast training speed [9]. Meanwhile, by constructing several small neural networks hierarchically, SELM achieves higher feature extraction capability and less memory occupation [10,11]. SELM has been introduced into many fields, such as classification and regression [12,13]. Our previous work [14], SELM has successfully introduced into the power flow analysis with cheap computational cost. However, the learning ability of SELM should be enhanced. Considering the nonconvexity and nonlinearity of the AC-OPF problem, further adjustments are required for SELM learning.

We summarize the main contributions of this paper in the following: a data-driven AC-OPF method assisted by Lagrange multipliers is proposed to enhance the learning ability of SELM for AC-OPF problems. Instead of incorporating the prior physical information into the neural networks algorithm, we developed a new neural network structure for the SELM learning based on the Lagrange multipliers. Specifically, we treat the Lagrange multipliers as an intermediate variable between the system power injections and the optimal solutions. The Lagrange multipliers can reflect the operating status of power systems, which contains the information of power system operating constraints stratification. Thanks to that, the additional features are introduced to the neural network structure, which significantly improves the learning ability of SELM.

2. Problem formulation

This section first shows the AC-OPF model. Then, the associated challenges of data-driven AC-OPF via SELM learning are discussed.

2.1. Physical model of the AC-OPF problem

The physical model of the AC-OPF problem is formulated as follows:

• Objective function

$$\min F = \sum_{i \in S_G} \left(a_{2i} PG_i^2 + a_{1i} PG_i + a_{0i} \right), \tag{1}$$

where S_G represents the sets of generator buses; PG_i is the active power output; a_{2i}, a_{1i}, and a_{0i} represent the quadratic, linear, and constant cost coefficients.

• Power balance constraint

$$PG_i - PD_i = V_i \sum_{j \in i} V_j (B_{ij} \sin\theta_{ij} + G_{ij} \cos\theta_{ij}) : \eta_{P,i}, i \in S_B, \tag{2}$$

$$QG_i - QD_i = V_i \sum_{j \in i} V_j (-B_{ij} \cos\theta_{ij} + G_{ij} \sin\theta_{ij}) : \eta_{Q,i}, i \in S_B, \tag{3}$$

where S_B represents the sets of all buses; QG_i is the reactive power output; PD_i and QD_i represent the active and reactive power injection, respectively; V_i are voltage magnitude; θ_i represent voltage phase angle; $\eta_{P,i}$ and $\eta_{Q,i}$ are the Lagrange multipliers of active and reactive power balance constraints.

• Transmission power flow constraint

$$PF_{ij} \le F_l : \eta_{PF,ij}^{\max}, \quad ij \in S_L, \tag{4}$$

$$PF_{ij} \leq F_l : \eta_{PF,ij}^{\min}, \quad ij \in S_L, \tag{5}$$

where S_L represents the sets of transmission lines; PF_{ij} is the transmission power flow from bus i to bus j; $\eta_{PF,ij}^{max}$ and $\eta_{PF,ij}^{\min}$ represent the Lagrange multipliers of the forward and backward transmission constraints.

- Generator active and reactive power output constraint

$$PG_i \leq \overline{PG_i} : \eta_{PG,i}^{\max}, \qquad i \in S_G, \tag{6}$$

$$PG_i \geq \underline{PG_i} : \eta_{PG,i}^{\min}, \qquad i \in S_G, \tag{7}$$

$$QG_i \leq \overline{QG_i} : \eta_{QG,i}^{\max}, \qquad i \in S_G, \tag{8}$$

$$QG_i \geq \underline{QG_i} : \eta_{QG,i}^{\min}, \qquad i \in S_G, \tag{9}$$

where $\eta_{PG,ij}^{max}$ and $\eta_{PG,ij}^{\min}$ represent the Lagrange multipliers of the generator upper and lower active power output constraint, respectively; $\eta_{QG,ij}^{max}$和$\eta_{QG,ij}^{\min}$ represent the Lagrange multipliers of the generator upper and lower reactive power output constraint, respectively.

- Voltage magnitude constraint

$$V_i \leq \overline{V_i} : \eta_{V,i}^{\max}, \quad i \in S_B \tag{10}$$

$$V_i \geq \underline{V_i} : \eta_{V,i}^{\min}, \quad i \in S_B \tag{11}$$

where $\eta_{V,i}^{max}$ and $\eta_{V,i}^{\min}$ represent the Lagrange multipliers of the upper and lower voltage magnitude constraint, respectively.

2.2. Problem statement

To overcome the massive hyperparameter adjustment of DNN, we developed a SELM network to learn the AC-OPF problem. The regression function of SELM is formulated in the following:

$$y_{SELM}(\boldsymbol{x}) = \boldsymbol{B}^T(\boldsymbol{x})\boldsymbol{\Upsilon} \tag{12}$$

where $\boldsymbol{B}(\boldsymbol{x})$ is the hidden layer output vector; $\boldsymbol{\Upsilon}$ represents the output weight vector. The core idea of SELM is to calculate $\boldsymbol{B}(\boldsymbol{x})$ and $\boldsymbol{\Upsilon}$ by iteratively solving the L2 regularization optimization problem in the following.

$$\min_{\boldsymbol{\Upsilon}^{(i)}} \left\{ f^{(i)} = \varsigma \left\| \mathbf{T} - \boldsymbol{B}^{(i)}\boldsymbol{\Upsilon}^{(i)} \right\|_2^2 + \left\| \boldsymbol{\Upsilon}^{(i)} \right\| \right\} \tag{13}$$

where $\boldsymbol{\Upsilon}^{(i)}$ and $\boldsymbol{B}^{(i)}$ are the output weight vector and hidden layer output vector in the ith iteration, respectively; ς represents a penalty factor, providing a tradeoff between the output weights norm and SELM training error. More details about the SELM can also be found in [9–11].

This paper develops a SELM network to learn the relationships between power injections and optimal solutions of AC-OPF problem, as can be seen in Fig. 1. The complicated AC-OPF mapping relationship can be represented by f: PD_i, QD_i $\rightarrow$ PG_i, QG_i, V_i, θ_i, F, which is the learning target of the SELM.

However, as can be seen from Eq. (12), the essence of SELM is a single-layer neural network with limited learning ability. Hence, the mapping relationship of the AC-OPF problem cannot be learned by the original SELM, which will be demonstrated in Section 4. Further adjustments are required for SELM learning.

3. Data-driven AC-OPF learning framework assisted by Lagrange multipliers

This section proposes a data-driven AC-OPF learning framework that is based on a Lagrange multiplier assisted method.

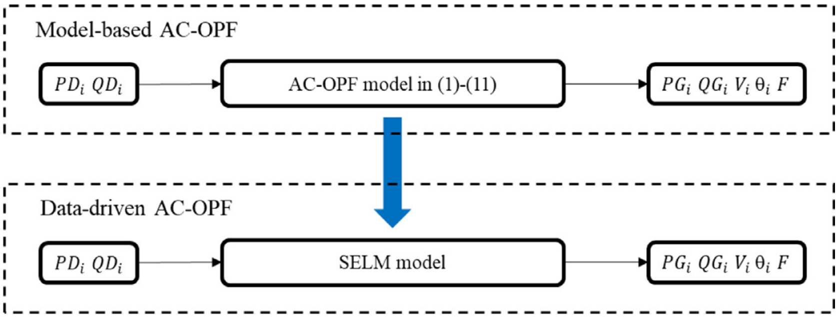

Fig. 1. SELM model for AC-OPF problem.

3.1. Lagrange multiplier assisted AC-OPF learning

The model-based OPF method can automatically produce Lagrange multipliers, which is able to evaluate the variation of the objective function (1) with respect to variation constraints (2)–(11). This allows us to obtain additional information to assist the AC-OPF learning. As a result, the learning ability of SELM can be improved by constructing a learning framework that contains the information of Lagrange multipliers.

However, as can be in the AC-OPF model (1)–(11), the number of Lagrange multipliers in an AC-OPF problem is large. Using all the Lagrange multipliers to assist the AC-OPF learning will inevitably enlarge the scale of the problem. Hence, the Lagrange multipliers that contain important features of AC-OPF should be selected. In this paper, we choose the Lagrange multipliers associated with the power balance constraint $\eta_{P,i}$, $\eta_{Q,i}$ to assist the AC-OPF learning, while other Lagrange multipliers are ignored in this paper.

The Lagrange multipliers $\eta_{P,i}$, $\eta_{Q,i}$ are important features of the AC-OPF problem, also known as the locational marginal price (LMP) in the electricity market [15]. The physical meaning of $\eta_{P,i}$ and $\eta_{Q,i}$ is the marginal cost to service the power injection next increment, which can reflect the allocation of energy in the optimal power operating point. Based on that, we construct a data-driven AC-OPF learning framework by adding Lagrange multipliers into the neural network to improve the learning performance, as shown in Fig. 2. However, as can be seen from Eq. (12), the learning ability of SELM is limited. Hence, the mapping relationship of the AC-OPF problem cannot be learned by the original SELM, which will be shown in the case study. To enhance the SELM learning ability, the AC-OPF learning task is divided into two stages in the proposed data-driven learning framework. The details procure are explained in the following:

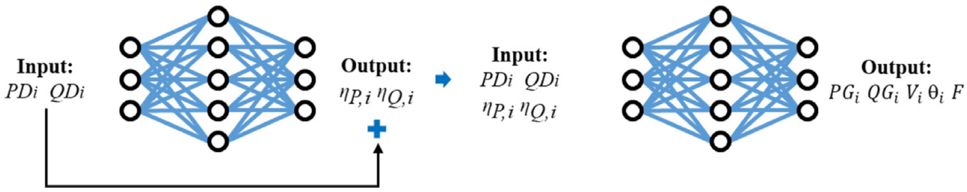

Fig. 2. SELM learning framework for AC-OPF problem.

Stage 1 (f : PD_i, QD_i $\rightarrow$ $\eta_{P,i}$, $\eta_{Q,i}$): learns the Lagrange multipliers of active and reactive power balance constraints. We introduce the Lagrange multipliers as intermediate variables to assist the SELM learning for the OPF problem. Through the historical power flow data PD_i, QD_i and $\eta_{P,i}$, $\eta_{Q,i}$ from the historical database, the additional information of Lagrange multipliers can be embedded into the neural network for SELM learning.

Stage 2 (f : PD_i, QD_i, $\eta_{P,i}$, $\eta_{Q,i}$$\rightarrow$$PG_i$, QG_i, V_i, θ_i, F): the learned Lagrange multipliers are combined with power injections as inputs of stage 2 to learn the AC-OPF solutions. The Lagrange multipliers cover the allocation of energy in the optimal power operating points, which is highly related to the AC-OPF solutions. Hence, it can perform significant efficiency for learning the AC-OPF solution.

The essence of the proposed SELM framework is to introduce additional Lagrange multipliers information to assist the AC-OPF learning.

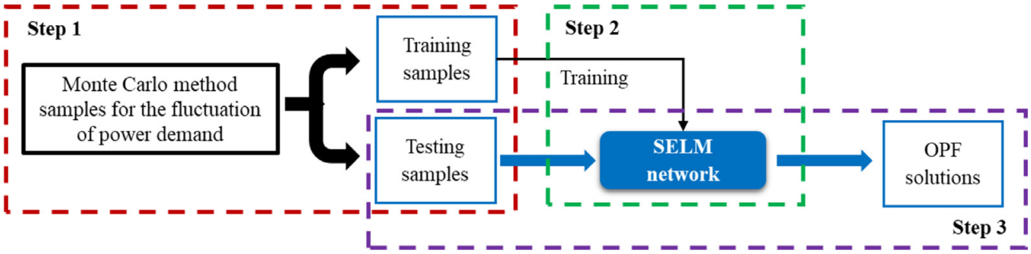

Fig. 3. Flowchart of the proposed method.

3.2. Proposed method flowchart

This paper constructs a data-driven AC-OPF learning framework to learn the nonlinear mapping relationship between the power injections and the optimal solutions. Fig. 3 shows the flowchart of the proposed method. It has four steps:

Step 1. Simulation of training samples: Training samples are required to train a SELM network for the AC-OPF problem. In this paper, the traditional interior-point algorithm and Monte Carlo simulation (MCS) are employed to obtain the training samples. The input of SELM are the system power injections PD_i, QD_i; the output of SELM are the OPF solutions PG_i, QG_i, V_i, θ_i, F. The Lagrange multipliers associated with the power balance constraint $\eta_{P,i}$, $\eta_{Q,i}$ also should be collected as the intermediate variables.

Step 2. Network training: Corresponding training samples are collected to train several SELM networks based on the proposed data-driven AC-OPF learning framework.

Step 3. Data-driven calculation: Input practice system power injections PD_i, QD_i to the SELM network trained in step 2. Then, the AC-OPF problem will be solved with low computational cost and high precision.

4. Case study

4.1. Case setting

In this section, the proposed approach is tested on several IEEE systems with the integration of renewable energies to illustrate its effectiveness. Specifically, wind power that follows the Weibull distribution and photovoltaic that follows the beta distribution. The load fluctuation rate and renewable penetration rate are shown in Table 1. The algorithm is implemented in MATLAB. All simulations are performed on a PC equipped with an Intel(R) Core(TM) i7-8700 CPU @ 3.20 GHz and 16 GB RAM.

Table 1. Uncertainties of cases.

	Renewable	Load
IEEE 39	13.43%	5%
IEEE 57	28.78%	10%
IEEE 118	24.75%	10%

The several methods are introduced for comparison:

M0: Monte Carlo simulation method, which can be seen as a benchmark;

M1: Deep neural network based on multilayer perceptron [16]

M2: Deep neural network method based on stacked denoised autoencoder [17];

M3: Original SELM;

M4: The proposed method.

The 10 000 training samples and 5000 testing samples are collected for the case study. Table 2 shows each data-driven method hyperparameter setting, which is adjusted based on the artificial experience and Ref. [14]. The data associated with the case study are given in [18].

Table 2. Each data-driven method hyperparameter settings.

Cases	Methods	Hyperparameter settings
IEEE 39-bus system	M1	Branch size = 200; learning rate = 0.0001; 200 fine-tuning epochs; 3 × 100 network
	M2	Branch size = 200; learning rate = 0.0001; 200 fine-tuning epochs; 3 × 100 network
	M3	2 epochs; 100 reduced hidden nodes; 1000 nodes
	M4	2 epochs; 100 reduced hidden nodes; 1000 nodes
IEEE 57-bus system	M1	Branch size = 300; learning rate = 0.0001; 300 fine-tuning epochs; 3 × 200 network
	M2	Branch size = 300; learning rate = 0.0001; 300 fine-tuning epochs; 3 × 200 network
	M3	2 epochs; 100 reduced hidden nodes; 1000 nodes
	M4	2 epochs; 100 reduced hidden nodes; 1000 nodes
IEEE 118-bus system	M1	Branch size = 500; learning rate = 0.0001; 500 fine-tuning epochs; 3 × 200 network
	M2	Branch size = 500; learning rate = 0.0001; 500 fine-tuning epochs; 3 × 200 network
	M3	2 epochs; 100 reduced hidden nodes; 1000 nodes
	M4	2 epochs; 100 reduced hidden nodes; 1000 nodes

This paper introduce an accuracy index χ to measure the SELM learning performance, which is the probability of the learning error less than the threshold *thr*, as shown in the following:

$$\chi = \Pr(\left|Y - \hat{Y}\right| \leq thr), \tag{14}$$

where Y and Ŷ represent the actual value and predicted value of the AC-OPF solutions, respectively. The threshold *thr* of *PG* and *QG* is set as 2 MW and 2 MVA, respectively; The threshold *thr* of *V* and θ is set as 0.001 p.u and 0.2°, respectively; The threshold *thr* of the objective function *F* is set as 100$.

4.2. Proposed method evaluation

The learning performance of M3 and M4 are compared in the IEEE 39-bus system to show the learning performance of the proposed method, as shown in Table 3.

Table 3. Learning performance of the proposed method in IEEE 39-bus system.

	Training time (s)	Testing time (s)	Accuracy index χ (%)				
			PG	*QG*	*V*	θ	*F*
M3	0.82	0.19	96.87	94.23	99.16	98.88	99.82
M4	2.17	0.52	99.62	99.57	99.67	99.82	99.98

Table 3 demonstrate that:

(1) The results of M3 demonstrate that the SELM direct learning is intractable, especially in PG and QG. This is because the AC-OPF model features are complicated that cannot be learned by a single-layer neural network.
(2) Comparing M3 with M4, the learning accuracy of *PG*, *QG*, *V*, θ, and *F* obtained from M4 increases by 2.75%, 5.34%, 0.51%, 0.94%, and 0.16%, respectively. It shows that the learning accuracy of all the AC-OPF solutions is improved because the additional Lagrange multipliers information is introduced into the SELM to assist the AC-OPF learning.

4.3. Comparison results

The learning performance of M0, M1, M2, and M4 are compared in the IEEE 39, 57, 118-bus system. The comparison results are shown in Table 4.

According to the learning performance shown in Table 4, we can see that:

(1) Among all the AC-OPF learning approaches, the proposed approach (M4) achieves the best learning performance.

Table 4. Comparisons of learning performance.

		Training time (s)	Testing time (s)	Accuracy index χ (%)				
				PG	*QG*	*V*	θ	*F*
IEEE 39-bus system	M0	–	63.27	–	–	–	–	–
	M1	77.95	0.31	89.02	81.35	99.37	86.98	54.54
	M2	28.03	0.05	76.03	62.74	95.97	67.53	26.26
	M4	2.17	0.52	99.62	99.57	99.67	99.82	99.98
IEEE 57-bus system	M0	–	158.83	–	–	–	–	–
	M1	100.76	0.33	92.70	81.69	94.92	89.65	31.80
	M2	66.54	0.11	80.55	68.16	83.84	73.45	16.80
	M4	3.32	0.73	99.23	99.14	97.79	99.74	99.92
IEEE 118-bus system	M0	–	322.82	–	–	–	–	–
	M1	136.79	0.37	91.81	97.15	97.86	81.52	22.90
	M2	136.55	0.13	82.34	89.46	91.83	55.27	10.06
	M4	4.26	1.02	99.86	99.99	99.94	100.00	96.02

(2) M1, M2, and M4 have acceptable testing times, which are much faster than the model-based AC-OPF method in M0. Especially, M0 requires hundreds of seconds to compute 5000 samples for the AC-OPF problem in a large system. Compared with that, the proposed method can obtain similar AC-OPF solutions with only one second.

(3) The training time of the DNN-based methods, i.e., M1 and M2, is much longer than the proposed method. This is due to the well-known back-propagation process of DNNs. Compared with DNNs-based methods (M1 and M2), the proposed method only needs a few seconds to obtain a trained natural network while achieving better learning accuracy for the AC-OPF problem. Besides, compared with DNN methods that require massive hyperparameters tuning, the proposed method can be easily applied to several IEEE systems while retaining high learning accuracy with few adjustments of hyperparameters.

5. Conclusion

A data-driven AC optimal power flow method assisted by Lagrange multipliers is proposed in this paper. The Lagrange multipliers associated with the power balance constraint are introduced to construct a SELM network for AC-OPF learning. Compared with existing DNN-based methods, the proposed method achieves the best AC-OPF learning performance, and it can be easily applied to several IEEE systems. The future work will be a focus on the data-driven AC-OPF method considering the topological changes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

Project supported by National Natural Science Foundation of China (52077016) and in part by Chongqing Graduate Scientific Research Innovation, China Project (CYB20015).

References

[1] Mohan TM, Nireekshana T. A genetic algorithm for solving optimal power flow problem. In: Proc. 3rd int. conf. electron. commun. aerosp. technol.. 2019, p. 1438–40.

[2] Reddy SS. Optimal power flow using hybrid differential evolution and harmony search algorithm. Int J Mach Learn Cybern 2018;10:1077–91.

[3] Yang Y, Yu J, Yang Z, Xiang M, Liu R. Fast calculation of probabilistic optimal power flow: A deep learning approach. In: 2019 IEEE power & energy society general meeting. PESGM, Atlanta, GA, USA; 2019, p. 1–5.
[4] Pan X, Zhao T, Chen M, Zhang S. DeepOPF: A deep neural network approach for security-constrained DC optimal power flow. IEEE Trans Power Syst 2021. early access.
[5] Huang W, Pan X, Chen M, Low SH. DeepOPF-V: Solving AC-OPF problems efficiently. IEEE Trans Power Syst 2022;37(1):800–3.
[6] Velloso A, Van Hentenryck P. Combining deep learning and optimization for preventive security-constrained DC optimal power flow. IEEE Trans Power Syst 2021;(2021). early access.
[7] Fioretto F, Mak TWK, Hentenryck PV. Predicting AC optimal power flows: Combining deep learning and Lagrangian dual methods. In: Proc. AAAI conf. artificial intelligence, Vol. 34. 2020, p. 630–7, 1.
[8] Chatzos M, Fioretto F, Mak TW, Van Hentenryck P. High-fidelity machine learning approximations of large-scale optimal power flow. 2020, arXiv: arXiv:2006.16356.
[9] Huang GB, Zhu QY, Siew CK. Extreme learning machine: Theory and applications. Neurocomputing 2006;70:489–501.
[10] Zhou H, Huang G, Lin Z, Wang H, Soh YC. Stacked extreme learning machines. IEEE Trans Cybern 2015;45(9):2013–25.
[11] Lei X, Yang Z, Yu J, Zhao J, Gao Q, Yu H. Data-driven optimal power flow: A physics-informed machine learning approach. IEEE Trans Power Syst 2021;36(1):346–54.
[12] Wong CM, Vong CM, Wong PK, Cao J. Kernel-based multilayer extreme learning machines for representation learning. IEEE Trans Neural Netw Learn Syst 2018;29(3):757–62.
[13] Li B, He Y. An improved resnet based on the adjustable shortcut connections. IEEE Access 2018;6:18967–74.
[14] Gao Q, Yang Z, et al. Model-driven architecture of extreme learning machine to extract power flow features. IEEE Trans Neural Netw Learn Syst 2020. early access.
[15] Liu H, Tesfatsion L, Chowdhury AA. Locational marginal pricing basics for restructured wholesale electricity markets. In: Proc. power energy soc. general meeting. 2009, p. 1–8.
[16] Brent RP. Fast training algorithms for multilayer neural nets. IEEE Trans Neural Netw 1991;2(3):346–54.
[17] Yang Y, Yang Z, Yu J, Xie K, Jin L. Fast economic dispatch in smart grids using deep learning: An active constraint screening approach. IEEE Internet Things J 2020;7(11):11030–40.
[18] Zhu Z. The data associated with this paper. Available on: https://figshare.com/articles/dataset/The_data_associated_with_the_paper_Data-Driven_Alternating_Current_Optimal_Power_Flow_A_Lagrange_Multipliers_Based_Approaches_/17704808.